

OPERATING SYSTEM & LINUX — QUICK NOTES

Based on the supplied handwritten notes — condensed into a maximum of 4 pages.

1. Operating System Overview

An Operating System (OS) is system software that manages computer hardware and provides services and an interface for users and applications.

Major OS responsibilities / topics

- Process Management: creates, schedules, and controls processes.
- Services: provides common system services to applications and users.
- Memory Management: allocates and manages RAM and virtual memory.
- Searching / File Management: helps locate, organize, and access files and resources.
- Package Management: downloads, installs, updates, and removes software packages.
- Security Management: protects systems using authentication, permissions, antivirus/security controls, etc.
- Scripting: automates administration and repetitive tasks.
- Extra Security: additional hardening, monitoring, access controls, and protection.
- Server Management: manages services, users, resources, and server operations.
- Troubleshooting: identifies and resolves OS, application, network, and hardware problems.

2. OS Components

The main components/tools noted are:

- OS / Kernel: core software that manages hardware and system resources.
- Shell (Command Line): command-based interface between the user and OS.
- Interpreter: executes instructions/scripts directly.
- Compiler: translates source code into machine/object code.
- Hardware: physical components such as CPU, memory, storage, and I/O devices.

3. Common OS Families / Examples

Family / OS	Examples / Notes
Linux	Red Hat family: Red Hat Enterprise Linux (RHEL), CentOS, Fedora. Debian family: Debian, Ubuntu, Kali, etc.
Windows	Microsoft desktop/server operating system.
macOS	Apple desktop operating system.
Unix	Unix and Unix-like operating systems.
Android	Mobile OS based on the Linux kernel.

4. Booting Process — Basic Idea

- Windows: firmware/boot manager loads the Windows boot components, then the OS kernel and required services/drivers are started.
- Linux: firmware loads a bootloader (commonly GRUB), which loads the Linux kernel and starts the system.
- macOS: firmware/boot process loads macOS components and the kernel, followed by system services.
- Unix / Android: follow platform-specific firmware/bootloader stages before loading the kernel and system services.

LINUX OVERVIEW & CONNECTIVITY

5. Linux Overview

Linux is a family of Unix-like operating systems built around the Linux kernel. The notes group common distributions into two broad families:

Family	Examples
Red Hat family	RHEL, CentOS, Fedora
Debian family	Debian, Ubuntu, Kali, etc.

Linux learning topics: connectivity, basic Linux commands, package management, security, scripting, services, storage, users/permissions, and troubleshooting.

6. Connecting to Linux from Windows

A Windows computer can connect to a Linux machine using a third-party tool such as PuTTY or a command-line client such as PowerShell / Windows Terminal.

Typical SSH connection:

- Protocol: SSH (Secure Shell)
- Transport protocol: TCP
- Default SSH port: 22
- The client connects to the Linux server's IP address/hostname and authenticates with a username/password or key.

Connection flow

Windows PC → PuTTY / PowerShell → SSH/TCP → Port 22 → Linux Server

SSH provides an encrypted remote shell session for administration.

7. Windows Remote Desktop (RDP)

For graphical remote access to Windows, the notes identify RDP (Remote Desktop Protocol).

- Protocol: RDP
- Transport: TCP
- Default RDP port: 3389
- Used for a graphical user interface (GUI) session to a Windows computer/server.

8. Linux ↔ macOS / Other Unix-like Systems

- macOS → Linux: SSH over TCP, normally port 22.
- Linux → macOS: SSH over TCP, normally port 22, when SSH access is enabled.
- Linux → Linux: SSH over TCP, normally port 22.
- Linux → Windows: SSH can be used when an SSH server is enabled; for GUI access, RDP may be used.

Simple server access model

User → Linux workstation/server → Linux server (1) / Windows server (2) / Mac server (3)

The exact protocol and port depend on the service being used.

SHELLS, COMMANDS & PACKAGE MANAGEMENT

9. What is a Shell in Linux?

A shell is a command-line interface (CLI) that allows a user to communicate with the operating system by typing commands. It can also execute scripts.

Shell	Full name / type	Common use
sh	Bourne shell	Basic Unix/Linux scripting and compatibility
bash	Bourne Again Shell	Common Linux shell; administration and scripting
zsh	Z shell	Interactive terminal and scripting; popular on modern systems
ksh	Korn shell	Enterprise/Unix scripting
csh	C shell	C-like syntax; historical Unix systems
tcsh	TENEX C shell	Enhanced C shell with interactive features
fish	Friendly Interactive Shell	Beginner-friendly interactive terminal
dash	Debian Almquist Shell	Lightweight shell and scripting
ash	Almquist shell	Embedded/minimal Unix-like systems

10. Basic Linux Commands

Command	Purpose
pwd	Shows the current working directory.
ls	Lists files and directories.
cd	Changes the current directory.
mkdir	Creates a directory.
touch	Creates an empty file or updates a file timestamp.
cp / mv	Copies / moves files or directories.
rm	Removes files; use carefully.
cat	Displays file contents.
grep	Searches text for matching patterns.
ps / top	Views running processes / system activity.
chmod / chown	Changes permissions / ownership.
ssh	Connects to a remote system securely.

11. Package Management

- A package manager downloads, installs, updates, and removes software packages and their dependencies.
- Debian/Ubuntu: APT is commonly used (for example, apt update, apt install package).
- RHEL/Fedora family: DNF is commonly used (for example, dnf install package).
- The handwritten notes specifically emphasize package downloading and installation.

12. Security & Administration

- Security management includes authentication, authorization, permissions, patching, firewalls, antivirus/security tools, and monitoring.
- Scripting can automate routine administration and troubleshooting.
- Server management covers services, users, resources, connectivity, and system health.

CLOUD CONCEPTS & REVISION SHEET

13. What is Cloud?

Cloud computing is the delivery of computing resources and services over a network, commonly the Internet. The notes describe the cloud as a collection of domains/resources.

Domains / areas listed in the notes

- Networking
- Database
- Storage
- Virtualization
- Containerization
- Security
- IAM (Identity and Access Management)
- Troubleshooting
- Automation
- Reverse engineering

14. Cloud Provider Terms from the Notes

Provider	Term used in the notes	General idea
AWS	Services	Cloud services/resources are consumed through AWS.
Microsoft Azure	Resources	Cloud resources are provisioned and managed through Azure.
Google Cloud	Engines	Cloud compute and other resources are provided through Google Cloud.

15. Quick Connectivity Reference

Connection	Common method	Protocol	Port
Windows → Linux	PuTTY / PowerShell / Windows Terminal	SSH over TCP	22
Windows → Windows	Remote Desktop	RDP over TCP	3389
Linux → Linux	SSH	SSH over TCP	22
macOS → Linux	SSH	SSH over TCP	22
Linux → macOS	SSH	SSH over TCP	22

16. Exam / Interview Revision Points

- OS: manages processes, memory, files, devices, security, and services.
- Shell: CLI used to communicate with the OS and run commands/scripts.
- SSH: secure remote shell; normally TCP port 22.
- RDP: Windows graphical remote desktop; commonly TCP port 3389.
- Linux families: Red Hat family and Debian family are major distribution families.
- Package management: use the distribution's package manager to install/update/remove software.
- Cloud: on-demand computing resources/services such as networking, storage, databases, virtualization, security, and automation.

One-line architecture: Hardware → Kernel/OS → Shell & System Services → Applications → User

Remote access: Client → Network → Protocol (SSH/RDP) → Port → Server → Authentication → Session